ATORVASTATIN

Myopathy risk

Contraindicated in pregnancy

CHEMICAL NAME (3R,5R)-7-[2-(4-fluorophenyl)-3-phenyl-4-(phenylcarbamoyl)-5-(propan-2-yl)-1H-pyrrol-1-yl]-3,5-dihydroxyheptanoic acid (calcium)

CLASS HMG-CoA reductase inhibitor (statin), high intensity

TRADE NAMES Lipitor · Sortis · Atorva · Torvast

THERAPEUTIC USES / INDICATIONS

- Primary and secondary prevention of cardiovascular disease (MI, stroke).
- Hypercholesterolaemia (raised LDL); mixed dyslipidaemia.
- Acute coronary syndrome (high-dose, early).
- Diabetes with CV risk; familial hypercholesterolaemia.

ADMINISTRATION

- **PO** once daily at any time of day (long half-life), with or without food.

DOSAGES

- **Primary prevention:** 20 mg once daily.
- **Secondary prevention / ACS:** 80 mg once daily.
- Range 10–80 mg.
- With clarithromycin, HIV PIs: avoid or max 20 mg; with ciclosporin: avoid.

PHARMACOKINETICS

- **Onset:** LDL falls within 2 weeks; max effect 4 weeks.
- Extensive first-pass metabolism by CYP3A4 (active metabolites).
- Excreted in bile. **t½:** 14 h (activity 20–30 h).

PHARMACODYNAMICS

- Inhibits **HMG-CoA reductase** (rate-limiting enzyme of cholesterol synthesis) in liver → ↑ hepatic LDL receptors → ↓ LDL-C (up to 50–60%).
- Modest ↓ triglycerides, ↑ HDL; plaque stabilisation, anti-inflammatory (pleiotropic) effects.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Myalgia (muscle aches).
- GI upset, headache.
- Raised liver enzymes.

ADVERSE EFFECTS (SERIOUS)

- Myopathy and rhabdomyolysis (dark urine, muscle weakness) — ↑ with high dose, interacting drugs, hypothyroidism, renal impairment.
- Hepatitis (rare).
- New-onset diabetes (small ↑).
- Immune-mediated necrotising myopathy (rare).

CONTRAINDICATIONS & CAUTIONS

- Active liver disease or unexplained persistent raised transaminases.
- Pregnancy and breastfeeding (stop 3 months before conception).
- Use with glecaprevir/pibrentasvir; tipranavir+ritonavir.

KEY INTERACTIONS

- **CYP3A4 inhibitors** (clarithromycin, itraconazole, HIV PIs), **ciclosporin, gemfibrozil:** ↑ myopathy.
- **Fusidic acid (systemic):** rhabdomyolysis; stop statin.
- **Grapefruit juice (large amounts):** ↑ levels.
- **Digoxin:** slight ↑ levels.

NURSING MANAGEMENT

- Baseline lipid profile, LFTs, CK (if symptoms), HbA1c.
- Repeat lipids and LFTs at 3 months.
- Ask about unexplained muscle pain or weakness; check CK.
- Review interacting drugs.
- Reinforce lifestyle measures.

PATIENT EDUCATION

- Take once daily every day; benefit is long-term.
- Report unexplained muscle pain, weakness or dark urine.
- Avoid large amounts of grapefruit juice.
- Stop and tell your doctor if pregnant.

KEY POINTS

- High-intensity statin — HMG-CoA reductase inhibitor.
- Myopathy/rhabdomyolysis; CYP3A4 interactions.
- 20 mg primary, 80 mg secondary prevention.
- Contraindicated in pregnancy.